

Group Assignment: Computer Networks and Network Operating Systems

Instructions:

Each group will research and present on their assigned topic. The presentation should focus on explanations with relevant examples without requiring practical demonstrations or the use of physical devices, except for **Groups 3, 7, and 8**, which will conduct practical presentations. Additionally, groups should submit a written report (3-5 pages) and a short quiz (3-5 questions) based on their topic.

Group Assignments:

Group 1: Basic Computer Network Design and Implementation

- Explain the key principles of designing a computer network.
- Describe network configuration methods and best practices.
- Provide examples of a simple network setup with basic security measures.

Group 2: Functions of Network Operating Systems vs. Other Operating Systems

- Define Network Operating Systems (NOS) and compare them with other operating systems.
- Explain the unique functions of NOS (e.g., centralized user management, file sharing, remote access).
- Provide examples of popular NOS such as Windows Server, Linux Server, and Cisco IOS.

Group 3: Installing and Configuring a Network Operating System (Practical Presentation)

- Demonstrate the installation process of a network operating system (e.g., Windows Server or Linux Server).
- Explain basic configurations such as user management, network settings, and security policies.
- Provide examples of commonly used network operating systems and their configurations.

Group 4: Installing and Configuring Common Network Servers (Choose SMTP OR FTP)

- Explain the role of either SMTP (email server) or FTP (file transfer server).

- Describe the installation and configuration process of the chosen server (e.g., Postfix for SMTP, vsftpd for FTP).
- Provide examples of real-world usage of the chosen server.

Group 5: Installing and Configuring Network Utilities (Choose DNS OR NFS)

- Explain the importance of either DNS (Domain Name System) or NFS (Network File System).
- Describe the installation and configuration of the chosen utility (e.g., BIND for DNS, NFS server for NFS).
- Provide examples of how the chosen utility is used in enterprise environments.

Group 6: Creating Different File Systems (ext4 & NTFS) and Managing UNIX File Types

- Explain the differences between ext4 and NTFS file systems.
- Describe how file systems are created, formatted, and managed.
- Provide examples of various UNIX file types (regular files, directories, symbolic links, etc.).

Group 7: Creating an Operating System Boot Disk with GRUB (Practical Presentation)

- Explain the function of a bootloader (GRUB) in operating systems.
- Describe how to create a bootable disk and configure multiple boot options.
- Provide examples of GRUB customization for different boot configurations.

Group 8: Configuring Dual Boot and Recovering a Non-Booting System (Practical Presentation)

- Explain dual boot and its advantages.
- Describe how to set up a system with two operating systems (e.g., Windows and Linux).
- Explain how to use a live Linux environment to recover a system that fails to boot.

Group 9: Managing Linux System Run Levels and Services

- Explain Linux run levels and how they affect system services.
 - Describe how to configure run levels and manage system services using commands (systemctl, service).
 - Provide examples of adjusting services to start at different run levels.
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Submission Requirements:

1. **Presentation:** Each group must prepare a 10-15 minute explanation-based presentation supported by conceptual diagrams and examples. Groups 3, 7, and 8 will conduct practical presentations.
2. **Written Report:** 3-5 pages covering their research, findings, and theoretical explanations.
3. **Quiz:** Prepare 3-5 questions to test classmates on the topic.